What is a Deployment in Kubernetes?

A **Deployment** is a Kubernetes object used to:

- **Create** and **manage ReplicaSets**
- Handle **rolling updates** and **rollbacks**
- Scale applications **up/down**
- Ensure **zero downtime** during updates
- Automatically **replace** unhealthy pods

Deployments are the **most commonly used** Kubernetes controller for managing stateless applications.

Deployment YAML Example (with Full Features)

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 3
  revisionHistoryLimit: 5
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 1
  selector:
  matchLabels:
    app: nginx
template:
  metadata:
    labels:
      app: nginx
  spec:
```

```
    containers:
    - name: nginx-container
      image: nginx:1.21
      ports:
      - containerPort: 80
      resources:
        limits:
          cpu: "200m"
          memory: "256Mi"
        requests:
          cpu: "100m"
          memory: "128Mi"
      readinessProbe:
        httpGet:
          path: /
          port: 80
        initialDelaySeconds: 5
        periodSeconds: 10
      livenessProbe:
        httpGet:
          path: /
          port: 80
        initialDelaySeconds: 15
        periodSeconds: 20
```

🔍 **Feature Breakdown:**

Feature	Description
replicas: 3	Run 3 copies of the pod.
strategy.rollingUpdate	Ensure zero-downtime updates (1 updated, 1 unavailable at most).
revisionHistoryLimit	Keep last 5 revisions for rollback.

Feature	Description
readinessProbe	Pod becomes available to service traffic **after** it is ready.
livenessProbe	Restarts pod if app becomes unhealthy.
resources	CPU/Memory limits and requests for resource control.

🛠️ **Practical Steps in Kubernetes**

1. Create the Deployment

kubectl apply -f nginx-deployment.yaml

2. Check Deployment Status

kubectl get deployments

kubectl get rs

kubectl get pods

3. Update the Image (Rolling Update)

kubectl set image deployment/nginx-deployment nginx-container=nginx:1.23

This triggers a **rolling update**, creating new pods while gracefully terminating old ones.

4. Check Rollout Status

kubectl rollout status deployment/nginx-deployment

5. Rollback to Previous Version

kubectl rollout undo deployment/nginx-deployment

6. Scale the Deployment

kubectl scale deployment/nginx-deployment --replicas=5

7. Pause and Resume Deployment

kubectl rollout pause deployment/nginx-deployment

kubectl rollout resume deployment/nginx-deployment

8. View Deployment History

kubectl rollout history deployment/nginx-deployment

📌 **Summary**

Action	Command
Create Deployment	kubectl apply -f

Action	Command
Update Image	kubectl set image
Rollback	kubectl rollout undo
Scale	kubectl scale
Monitor Status	kubectl rollout status
Pause/Resume	kubectl rollout pause/resume